ANNUAL REPORT PORTAL

Sprint 1 — Day-by-Day Project Documentation

Streamlining Academic and Institutional Performance Tracking through Digital Cloud Infrastructure

Timeline: June 01 – June 13, 2026 | **Version:** 1.0 | **Scope:** Mini-Project Submission (IT)

Day 1: Project Title Finalization

Date: **June 01, 2026** | Deliverable: **Approved Project Title**

1.1 Chosen Project Title

Annual Report Portal

1.2 Why This Title Was Chosen

The title was finalized after evaluating alternative concepts like a Personalized Career Advisor. The title 'Annual Report Portal' clearly identifies the system's core utility—centralizing data collection and generating institutional audits. It is brief, clear, and perfectly tailored for a 3rd-year Information Technology mini-project submission.

1.3 Project Abstract

The Annual Report Portal is an intelligent, web-based centralized repository designed to automate the compilation, analysis, and generation of an institution's annual performance reports. Traditionally, academic departments compile metrics manually, leading to data inconsistencies and delayed submissions.

This system provides a seamless platform where distinct institutional nodes (Departments, Placement Cells, R&D Wings, and Exam Cells) can securely log in and upload seasonal metrics. The platform aggregates these diverse data points using a structured relational backend, enabling administrators to generate comprehensive annual analytics reports instantly. Built using a React.js frontend interface and a robust relational data model, it eliminates manual friction and optimizes administrative auditing.

1.4 Scope

- Covers key institutional modules: Academic Performance, Placements, R&D, and Infrastructure.
- Provides granular access controls for Department Heads, Cell Coordinators, and System Administrators.
- Enables dynamic export configurations (such as PDF generation) for external regulatory audits.
- Focuses strictly on workflow tracking, data persistence, and report generation; external third-party ERP payroll integrations are excluded from this version.

1.5 Deliverable Summary — Day 1

Item	Status
Project title finalized and approved	Completed — June 01, 2026
Abstract drafted	Completed — June 01, 2026
Scope boundaries defined	Completed — June 01, 2026

Day 2: Requirement Gathering

Date: **June 02, 2026** | Deliverable: **Problem Statement**

2.1 Problem Statement

Educational and corporate institutions face immense challenges when compiling comprehensive performance metrics for their annual cycles. Data points remain siloed across individual spreadsheets, local drives, and disconnected department registries. Manually consolidating this information results in severe structural inefficiencies, clerical duplication errors, and a lack of real-time visibility for administrative authorities. There is an immediate requirement for an automated cloud pipeline that streamlines multi-tenant data input and builds clean, unified, and instantly auditable annual reports.

2.2 Functional Requirements

- **FR-01 (User Registration & Authentication):** System must provide secure role-based registration and login pipelines for users (Faculty, Admin, Editors).
- **FR-02 (Data Intake Forms):** Contextual forms for entering department statistics (e.g., student pass percentages, placement statistics, publication registries).
- **FR-03 (Automated Report Aggregation):** Core engine must dynamically pull stored parameters to assemble comprehensive, aggregated annual reports.
- **FR-04 (File Processing System):** Support for uploading supporting validation materials (e.g., signed placement receipts or certificates).
- **FR-05 (Audit Logs & Trackers):** History log displaying past edits, timestamp records, and identifying which user handled specific metric updates.

2.3 Non-Functional Requirements

- **NFR-01 (Performance):** Report consolidation and spreadsheet aggregation operations must execute within 3 seconds under typical loads.
- **NFR-02 (Security):** Encrypted user credential storage with secure session tracking using JavaScript Web Tokens (JWT).

- **NFR-03 (Usability):** A clean, modular user interface designed using accessible navigation principles optimized for both desktop and tablet screens.
- **NFR-04 (Scalability):** Relational architecture must handle concurrent input tasks across multi-department profiles without transaction blocks.

Day 3: Objective Definition

Date: **June 03, 2026** | Deliverable: **Project Objectives**

3.1 Primary Objective

To design and implement a web-based repository infrastructure that automates structural data gathering across institutional entities, simplifying administrative workflows by dynamically processing and exporting comprehensive annual performance audits.

3.2 Specific Objectives

- Configure a fully normalized relational schema to map multi-department parameters without redundancy.
- Build a modular, secure REST API backend using robust environment management principles.
- Develop an interactive React.js interface with dedicated dashboard workspaces specialized by user access privileges.
- Integrate structured report compiling engines capable of outputting production-ready document models.

Day 4: User & Module Identification

Date: **June 04, 2026** | Deliverable: **Module List**

4.1 Identified Users

- **System Administrator:** Oversees whole portal access control, initiates annual tracking parameters, and reviews compiled global reports.
- **Department Editor (HOD/Coordinator):** Enters specific operational details, uploads confirmation documents, and tracks their own department's submissions.

4.2 System Modules

- **M-01 (Auth Pipeline Module):** Standardizes validation protocols, maps user sessions, and controls role-based route permissions.
- **M-02 (Intake Panel Module):** Collects raw metrics using structured UI fields, validating numeric ranges before committing to storage.
- **M-03 (Report Core Engine):** Extracts multi-table data strings to synthesize institutional summary tables automatically.

- **M-04 (Audit History Tracker):** Keeps historical logs of data revisions to preserve database integrity and transparency.

Day 5: Use Case Diagram Preparation

Date: **June 05, 2026** | Deliverable: **CRUD APIs**

5.1 Use Cases & Structural Actions

- **UC-01 (Profile Creation):** New operational accounts built with distinct institutional access tokens.
- **UC-02 (Submit Report Data):** Users push current annual data parameters directly to live backend collections.
- **UC-03 (Compile Master Audit):** Administrative trigger to aggregate entries into an output-ready document summary.

5.2 Core API Endpoint Layout

- `POST /api/auth/login` → Session setup and verification token payload delivery.
- `GET /api/reports/summary` → Administrative query to fetch combined data points.
- `POST /api/metrics/submit` → Write operation to update specific data records.
- `PUT /api/metrics/update/:id` → Safely modifies verified data instances.

Day 6: Database Requirement Analysis

Date: **June 06, 2026** | Deliverable: **Table List**

6.1 Data Architecture Overview

A relational framework (such as MySQL or PostgreSQL) is essential to preserve the structural linkages between separate institutional components. This guarantees referential tracking so each submitted data record connects explicitly to an authenticated user account and a designated academic category.

6.2 Identified Tables

- **users:** Primary user account schemas (User ID, name, role, encrypted credentials).
- **department_metrics:** Captures quantitative inputs (Pass rates, research points).
- **uploaded_docs:** Maps paths for confirmation files attached during submission.
- **compiled_reports:** Persists structural pointers of finalized admin master audits.

Day 7: ER Diagram Design

Date: **June 07, 2026** | Deliverable: **ER Diagram**

7.1 Textual Relationship Architecture

```
┌───────────────┐ ┌────────────────────────┐ │ USERS │ │ DEPARTMENT_METRICS │
│───────────────│ │────────────────────────│ │ PK user_id │ 1 ──────── M│ PK
metric_id │ │ username │ │ FK user_id │ │ role_tag │ │ academic_year │
└───────────────┘ │ raw_data_values │ │ └────────────┬────────────┘ │ 1 │ 1 │ │ M │ M
┌───────────────┐ │ │ COMPILED_REPORTS │ │───────────────│ │ │ PK report_id │ M
───────────────────────┘ │ FK admin_id │ │ export_url│ └───────────────┘
```

Day 8: Database Schema Creation

Date: **June 08, 2026** | Deliverable: **SQL Schema**

8.1 SQL Schema Blueprint

```
CREATE DATABASE IF NOT EXISTS annual_report_portal_db;
USE annual_report_portal_db;

CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role_tag ENUM('admin', 'editor') DEFAULT 'editor',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE department_metrics (
  metric_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  academic_year VARCHAR(20) NOT NULL,
  student_pass_rate FLOAT NOT NULL,
  placement_count INT NOT NULL,
  publications_count INT NOT NULL,
  submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

CREATE TABLE compiled_reports (
  report_id INT AUTO_INCREMENT PRIMARY KEY,
  admin_id INT NOT NULL,
  report_title VARCHAR(150) NOT NULL,
  generated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (admin_id) REFERENCES users(user_id) ON DELETE RESTRICT
);
```

Day 9: UI Wireframe Design

Date: **June 09, 2026** | Deliverable: **Page Layouts**

9.1 Core Layout Paradigms

- **Data Integrity Focus:** Grid-based telemetry alignments ensuring information blocks match clean ledger styling.
- **Clear Action Mapping:** Full-width form submit actions paired with interactive validation status indicators.

Day 10: Login & Dashboard UI Design

Date: **June 10, 2026** | Deliverable: **UI Screens**

10.1 Interface Elements Layout

```
┌──────────────────────────────────────────────────────┐ │ [Portal Logo] Annual
Report Hub [Alerts] [User ▾] │
├──────────────────────────────────────────────────────┤ │ Welcome Back, Editor! 📍
Dept of IT │ │ │ │ ┌────────────────┐ ┌─────────────────┐ │ │ │ Submissions │ │
Current Cycle │ │ │ │ Verified │ │ 2025-2026 │ │ │ └─────────────────┘
└──────────────────┘ │ │ │ │ [ + START NEW METRIC SUBMISSION ] │ │ │
└──────────────────────────────────────────────────────┘
```

Day 11: Navigation & Form Design

Date: **June 11, 2026** | Deliverable: **UI Prototype**

11.1 Route Mapping Schema

- `/login` → Public access route for authentication validation.
- `/dashboard` → Main console tracking verification queues and logs.
- `/submit-metrics` → Ingestion panel built with conditional form layouts.
- `/admin/reports` → Restricted workspace for administrative data exports.

Day 12: Design Review

Date: **June 12, 2026** | Deliverable: **Design Approval**

12.1 Review Summary

All structural blueprints, data schemas, API routes, and interface frameworks were evaluated. The design successfully meets modern web standards, role-separation guidelines, and data validation rules, granting official clearance to move to the environment deployment stage.

Status: Design Phase Approved and Signed Off by Review Committee.

Day 13: Frontend Environment Setup

Date: **June 13, 2026** | Deliverable: **React Project Setup**

13.1 Setup Tasks Executed

- Initialized React development build targets using fast Vite configurations.
- Set up utility-first styles via Tailwind CSS configuration frameworks.
- Created organized component directories (`/components`, `/pages`, `/context`).
- Configured Axios routing rules and initialized local Git source control tracking.